

CO2Emissions Analysis Report

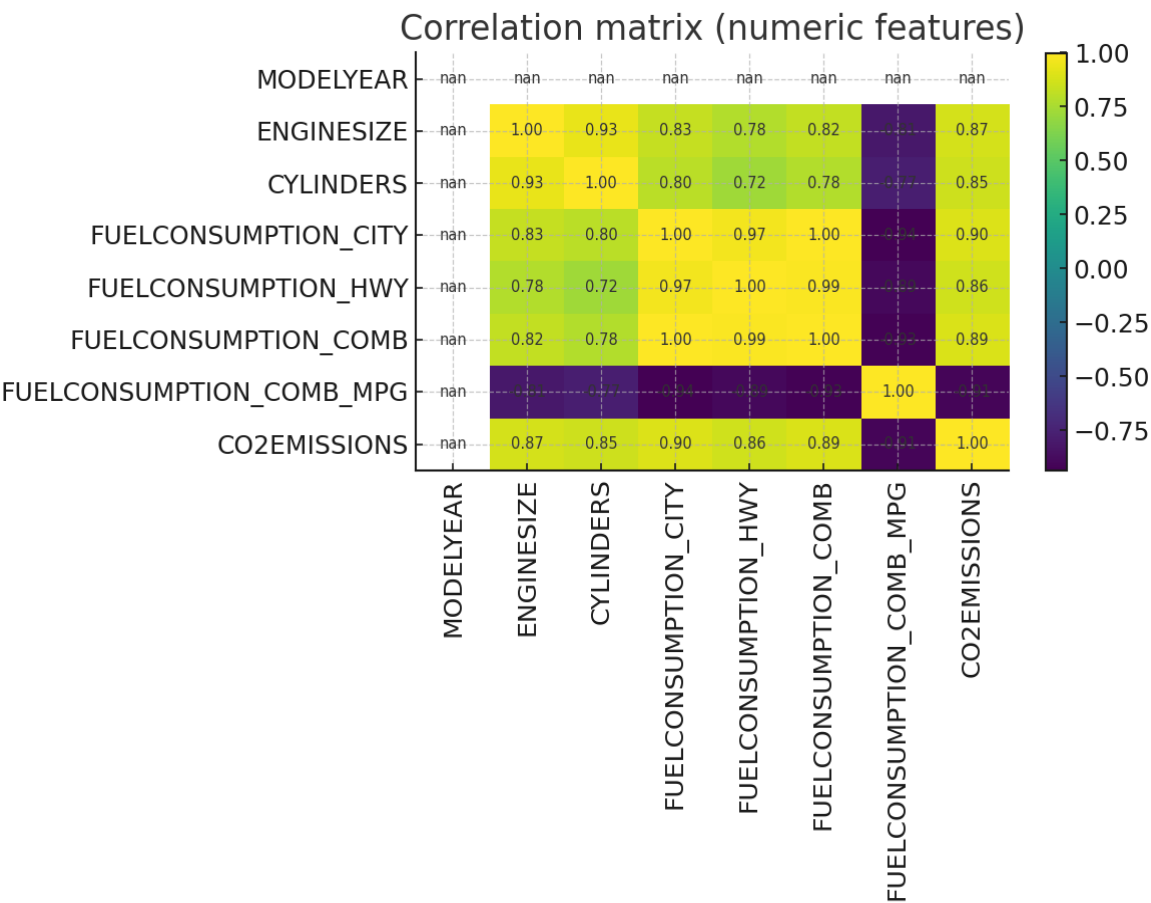
Dataset: FuelConsumptionCo2.csv

Dataset rows: 1067 Columns: 13
Mean CO2EMISSIONS: 256.23 g/km

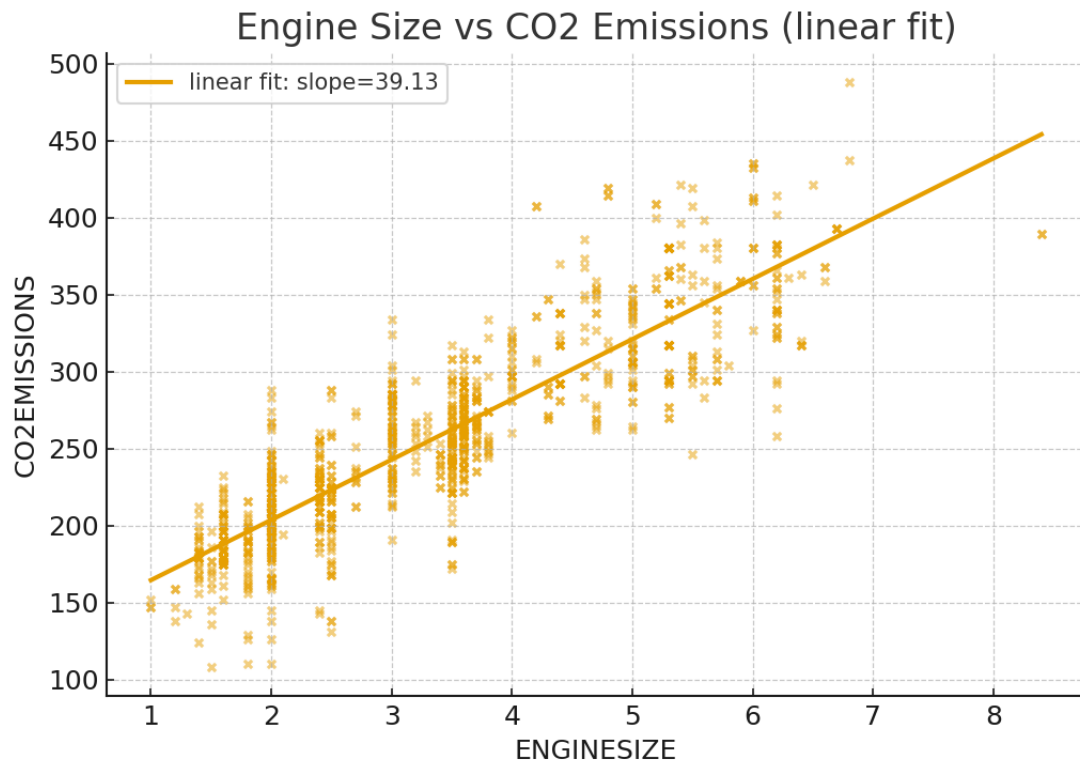
Top features correlated with CO2

Feature	Correlation with CO2
FUELCONSUMPTION_COMB_MPG	-0.906
FUELCONSUMPTION_CITY	0.898
FUELCONSUMPTION_COMB	0.892
ENGINE_SIZE	0.874
FUELCONSUMPTION_HWY	0.862
CYLINDERS	0.850

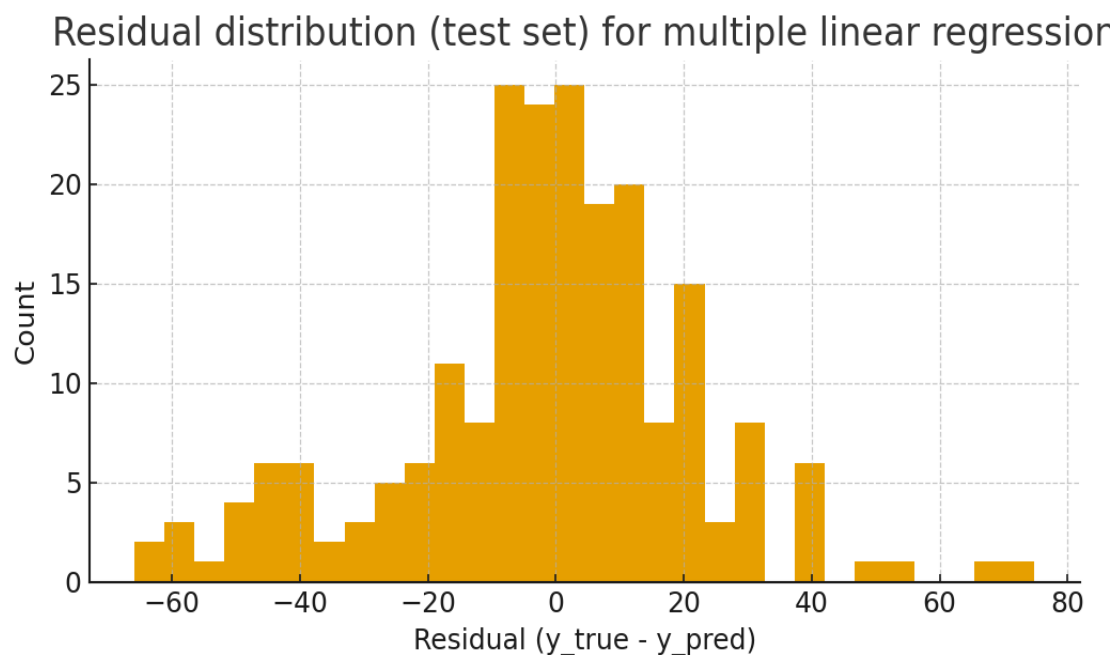
Correlation Matrix Heatmap



Engine Size vs CO2 Emission



Residual distribution for multi-feature linear regression



Modeling summary & interpretation

We fit simple linear baselines to explain CO2 emissions. A single-variable linear model using engine size explains a substantial portion of variance ($R^2 \approx 0.76$). Adding combined fuel consumption as a second feature improves performance ($R^2 \approx 0.87$; RMSE reduces from ≈ 31 g/km to ≈ 23 g/km). The coefficients are interpretable: they quantify the expected change in g/km CO2 per unit increase in the predictor. Residuals are centered near zero but show spread and a few outliers, indicating some remaining non-linear effects or data heterogeneity.

Recommendations & next steps

1. Use cross-validation and regularization (Ridge/Lasso) to stabilize coefficients and improve generalization.
2. Try low-degree polynomial features or interaction terms if residual plots show curvature.
3. Compare with tree-based models (RandomForest, XGBoost) for potential accuracy gains.
4. Investigate outliers and consider robust regression or target transformation if needed.